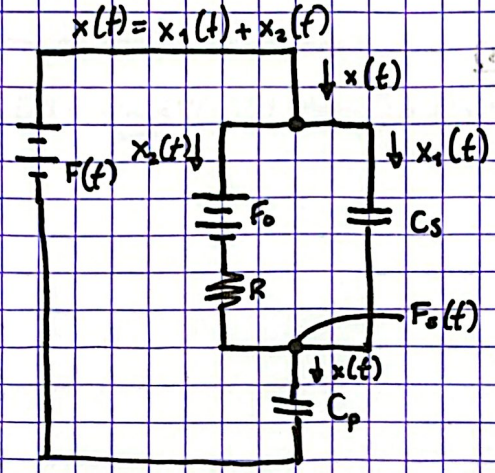
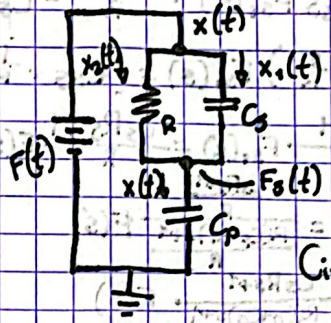
Diego Macías Armenta

Modelado

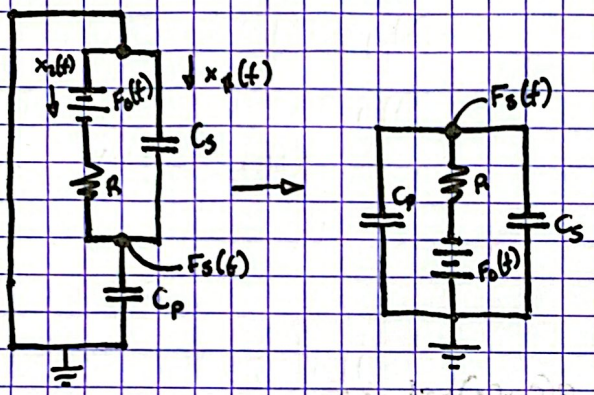
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Práctica 3: Sistema musculoesquelético





Circuito 1



Circuito 2

Análisis matemático

Debido a que el sistema contiene 2 fuentes de voltaje, se debe aplicar el principio de superposición.

Circuito 1 Apagando $f_0(t)$

$$x(t) = x_1(t) + x_2(t)$$

$$x(t) = C_p \frac{dF_{s1}(t)}{dt}$$

$$x_1(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$$

$$x_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p s F_{s1}(s) = C_s s [F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) = C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) + C_s s F_{s1}(s) + \frac{F_{s1}(s)}{R} = C_s s F(s) + \frac{F(s)}{R}$$

Norma

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Modelado
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$$C_p s F_{s_1}(s) + C_s s F_{s_1}(s) + \frac{F_{s_1}(s)}{R} = C_s s F(s) + \frac{F(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s_1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_p+C_s)s+1}{R} F_{s_1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s_1}(s) = \frac{\frac{C_s R s + 1}{R}}{\frac{R(C_p+C_s)s+1}{R}} F(s)$$

$$F_{s_1}(s) = \frac{C_s R s + 1}{R(C_p+C_s)s+1} F(s)$$

Circuito 2 Apagando $F(t)$

Recordando que $F_o(t) = aF(t)$

$$-aF(t) = R_x(t) + \frac{1}{C_p+C_s}\int x(t)dt$$

$$F_{s_2}(t) = \frac{1}{C_p+C_s}\int x(t)dt$$

$$-aF(s) = R_x(s) + \frac{x(s)}{(C_p+C_s)s}$$

$$F_{s_2}(s) = \frac{x(s)}{(C_p+C_s)s}$$

$$F(s) = -\frac{1}{a}\left(R + \frac{1}{(C_p+C_s)s}\right)x(s) = -\frac{1}{a}\frac{R(C_p+C_s)s+1}{(C_p+C_s)s}x(s)$$

$$\frac{F_{s_2}(t)}{F(t)} = \frac{\frac{x(s)}{(C_p+C_s)s}}{-\frac{1}{a}\frac{R(C_p+C_s)s+1}{(C_p+C_s)s}x(s)} = -\frac{a}{R(C_p+C_s)s+1}$$

$$F_{s_2}(t) = -\frac{aF(s)}{R(C_p+C_s)s+1}$$

Función de transferencia

$$F_s(s) = F_{s_1}(s) + F_{s_2}(t) = \frac{C_s R s + 1}{R(C_p+C_s)s+1} F(s) - \frac{aF(s)}{R(C_p+C_s)s+1}$$

$$F_s(s) = \frac{C_s R s + 1 - a}{R(C_p+C_s)s+1} F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R s + 1 - a}{R(C_p+C_s)s+1}$$

Valores:

Parámetro	Control	Caso
$F(t)$	1 V	1 V
a	0.25	0.25
C_s	10 µF	10 µF

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